Agromind MVP Models

Novel Smart Aquaculture and Agriculture System based on new AI Techniques

Agromind is an intelligent AI-powered platform designed to enhance aquaculture and agriculture productivity using advanced Machine Learning, Deep Learning, Time-Series Forecasting, Computer Vision, Optimization, and Large Language Models (LLMs). The system is structured into domain-based intelligent modules, where each domain contains specialized AI models designed to solve real-world farming and aquaculture challenges

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1. Aquaculture Intelligence Module

The Aquaculture Intelligence Module focuses on fish and shrimp farming optimization, disease management, growth forecasting, water monitoring, feeding optimization, stock estimation, and economic forecasting.

Below are the selected MVP models, completely separated for Fish and Shrimp operations to ensure maximum accuracy for each species.

A. Fish Intelligence Models (6 Models)

1. **Fish Disease Detection + Localization**

   - **Model:** U-Net / CNN-Based Classification
   - **Input:** Fish RGB image
   - **Output:** Disease Class + Confidence Score + Segmented infected area
   - **Function:** Early disease detection from pond images and measurement of the infected region for treatment planning.

2. **Fish Disease Severity Assessment**

   - **Model:** Regression-Based Severity Estimation / U-Net
   - **Input:** Fish image
   - **Output:** Severity Percentage (%) / Severity Level
   - **Function:** Quantify disease progression level to guide interventions.

3. **Fish Growth Prediction**

   - **Model:** Deep Learning Growth Model (LSTM/Transformer Time-Series)
   - **Input:** Growth history + Water parameters
   - **Output:** Predicted biomass / weight
   - **Function:** Long-term growth planning.

4. **Fish Feeding Recommendation**

   - **Model:** ML-Based Feed Optimization (XGBoost / Random Forest Regression)
   - **Input:** Fish weight + Water conditions
   - **Output:** Feed Quantity Recommendation
   - **Function:** Optimize feed usage to reduce waste and cost.

5. **Fish Counting (Stock Estimation)**

   - **Model:** YOLO-Based Detection
   - **Input:** Pond Image / Video
   - **Output:** Fish Count + Bounding Boxes
   - **Function:** Stock estimation automation and high-accuracy stock tracking.

6. **Fish Harvest Time Prediction**

   - **Model:** Harvest Forecast Model (XGBoost/LSTM)
   - **Input:** Growth + Market Price + Weather
   - **Output:** Optimal Harvest Date
   - **Function:** Maximize profit by timing the market correctly.

B. Shrimp Intelligence Models (6 Models)

1. **Shrimp Disease Detection + Localization**

   - **Model:** U-Net / CNN-Based Classification
   - **Input:** Shrimp RGB image
   - **Output:** Disease Class + Confidence Score + Segmented infected area
   - **Function:** Early disease detection from pond images and measurement of the infected region.

2. **Shrimp Disease Severity Assessment**

- **Model:** Regression-Based Severity Estimation / U-Net
- **Input:** Shrimp image
- **Output:** Severity Percentage (%) / Severity Level
- **Function:** Quantify disease progression level for targeted treatment.

3. **Shrimp Growth Prediction**

- **Model:** Deep Learning Growth Model (LSTM/Transformer Time-Series)
- **Input:** Growth history + Water parameters
- **Output:** Predicted biomass / weight
- **Function:** Long-term growth planning.

4. **Shrimp Feeding Recommendation**

- **Model:** ML-Based Feed Optimization (XGBoost / Random Forest Regression)
- **Input:** Shrimp weight + Water conditions
- **Output:** Feed Quantity Recommendation
- **Function:** Optimize feed usage.

5. **Shrimp Counting (Stock Estimation)**

- **Model:** YOLO-Based Detection
- **Input:** Pond Image / Video
- **Output:** Shrimp Count + Bounding Boxes
- **Function:** Automation of stock estimation for shrimp populations.

6. **Shrimp Harvest Time Prediction**

- **Model:** Harvest Forecast Model (XGBoost/LSTM)
- **Input:** Growth + Market Price + Weather
- **Output:** Optimal Harvest Date
- **Function:** Maximize profit by determining the best time to pull the stock.

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2. Agriculture Intelligence Module

The Agriculture Intelligence Module focuses on crop monitoring, yield prediction, irrigation optimization, soil analysis, disease detection, economic forecasting, and farm-level decision support.

1. **Crop Yield Prediction**

- **Model**: LightGBM
- **Input**: Weather data, Soil data, Crop info, Farm management data, and Location
- **Output**: Predicted Yield (tons/hectare)
- **Function**: Learns from past farming conditions to estimate seasonal crop production per unit area.

2. **Crop Disease & Weed Detection**

- **Model**: YOLOv8-seg (Object Detection + Segmentation)
- **Input**: Clear crop/leaf images
- **Output**: Segmentation Mask, Bounding Box, Disease/Weed Class Label, and Affected Area Percentage
- **Function**: Detects diseased areas and weeds directly from field images to enable precision spraying and early treatment.

3. **Irrigation Recommendation Management**

- **Model**: LightGBM
- **Input**: Soil moisture, Weather data, Crop stage, and Irrigation history
- **Output**: Irrigation Decision, Water Amount, and Schedule recommendation
- **Function**: Recommends whether to irrigate, how much water to use, and when to water next to optimize water usage.

4. **Harvest Optimal Time**

- Model: GRU (Gated Recurrent Unit)
- Input: Crop growth data, Weather, Soil data, and Historical harvest patterns
- Output: Optimal recommended harvest date
- Function: Predicts the exact time frame to harvest crops to maximize quality, yield, and profit.

5. **Crop Recommendation System**

- Model: Random Forest
- Input: Soil properties (NPK, pH), Weather/Climate data, and Location
- Output: Best Recommended Crop and Suitability Score
- Function: Analyzes soil and climate to suggest the most profitable and biologically suitable crop to plant.

6. **Fertilizer Optimization**

- Model: LightGBM
- Input: Soil nutrients, Crop type, Weather, and Expected yield target
- Output: Recommended Fertilizer Amount and Type
- Function: Provides precise nutrient application plans to reduce costs and boost sustainable growth.

7. **Ripeness Detection**

- Model: EfficientNetB4
- Input: RGB crop/fruit images
- Output: Ripeness stage classification (Unripe / Semi-Ripe / Ripe / Overripe)
- Function: Visually analyzes crops or fruits to determine their ripeness stage and optimize harvest timing.

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3. Weather & Environmental Intelligence Module

The Weather & Environmental Intelligence Module focuses on climate forecasting, environmental risk modeling, anomaly detection, resource stress prediction, and sustainability analytics for agriculture and aquaculture systems.

1. **Short-Term Weather Prediction**

- **Model**: LSTM / GRU
- **Input**: Historical temperature, humidity, rainfall
- **Output**: 1-7 day forecast
- **Function**: Operational planning

2. **Rainfall Intensity Prediction**

- Model: XGBoost / LSTM
- Input: Pressure, humidity, temperature
- Output: Rainfall amount (mm)
- Function: Irrigation planning

3. **Drought Risk Index**

- **Model**: Regression / ML
- **Input**: Rainfall deficit + Soil moisture
- **Output**: Drought severity score
- **Function**: Early stress detection

4. **Heat Wave Prediction & Stress Index**

- **Model**: LSTM / Transformer / Regression
- **Input**: Temperature time-series + Humidity
- **Output**: Heatwave probability + Stress index value
- **Function**: Crop & fish protection + Environmental impact scoring

5. **Flood Probability Model**

- **Model**: Random Forest / LSTM
- **Input**: Rainfall + Terrain + River levels
- **Output**: Flood risk score
- **Function**: Disaster mitigation

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4. Energy & Smart Resource Optimization Module

The Energy & Smart Resource Optimization Module focuses on intelligent energy forecasting, pump scheduling, renewable integration, cost minimization, and multi-objective optimization across agriculture and aquaculture systems. This module ensures sustainable operation while minimizing cost and maximizing productivity.

1. **Farm Energy Consumption Prediction**

- **Model**: LSTM/GRU
- **Input**: Historical energy usage + Operational logs
- **Output**: Future energy consumption (kWh)
- **Function**: Load planning

2. **Pump Operation Optimization**

- **Model**: Reinforcement Learning (DQN/PPO)
- **Input**: Water level + Soil moisture + Energy price
- **Output**: Pump ON/OFF schedule
- **Function**: Reduce energy cost

3. **Solar Power Generation Forecast**

- **Model**: LSTM/TFT
- **Input**: Solar radiation + Weather forecast
- **Output**: Predicted solar output (kW)
- **Function**: Renewable planning

4. **Smart Aerator Control (Aquaculture)**

- **Model**: Reinforcement Learning
- **Input**: DO levels + Biomass + Energy price
- **Output**: Aerator schedule
- **Function**: Maintain oxygen with minimal power

5. **Predictive Maintenance Model**

- **Model**: LSTM/Classification
- **Input**: Vibration + Temperature + Usage
- **Output**: Failure probability
- **Function**: Reduce downtime

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5. LLM & Cognitive Intelligence Layer (1 Model)

The LLM & Cognitive Intelligence Layer acts as the central reasoning engine of the Agromind platform. It integrates all AI domain outputs and transforms them into actionable, explainable, and strategic decisions.

1. **Conversational, Advisory & Reasoning Engine**

- **Model**: Fine-Tuned Lightweight LLM (e.g., LLaMA-3-8B / Mistral-7B) + RAG (Retrieval-Augmented Generation)
- **Input**: User natural language query + System data (yield, water, weather, energy, and ML model predictions)
- **Output**: Context-aware recommendations, actionable steps, and grounded responses with factual support
- **Function**: Provides real-time domain-aware advisory support, translating raw ML predictions into human-understandable explanations.

Models Distribution by Module

Module	Number	Main AI Types Used	Core Purpose
Aquaculture Intelligence	12	CNN, LSTM, XGBoost, Regression, Classification	Fish health monitoring, water quality prediction, feeding optimization
Agriculture Intelligence	7	CNN, Random Forest, Gradient Boosting, Time-Series	Crop disease detection, yield prediction, irrigation optimization
Weather & Environmental Intelligence	5	LSTM, ARIMA, Prophet, Regression	Weather forecasting, environmental risk detection
Energy & Smart Resource Optimization	5	Optimization Algorithms, ML Regression, Forecasting	Solar prediction, energy load forecasting, smart irrigation control
LLM & Cognitive Intelligence Layer	1	LLM, RAG, NLP, Prompt Engineering	Advisory system, reasoning, smart reporting
Total AI Models	**30 Models**	Hybrid AI Architecture	Fully Integrated Smart Agro-Aqua Platform